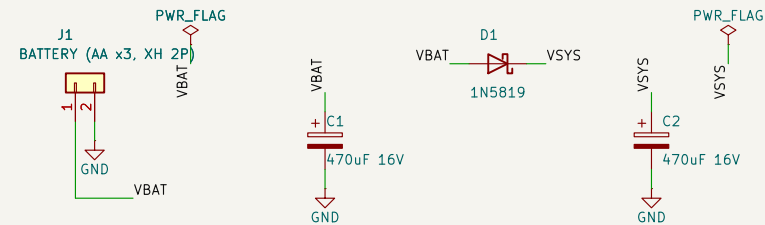


Linetracer2 Lite : low-cost version (Pico without pin headers, 3 sensors, 1 button, 1 LED)

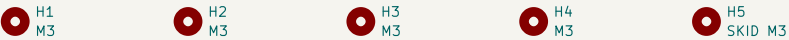
Power: AA x3 (alkaline 4.5V / NiMH 3.6V) -> VBAT (motors) ; VBAT -> D1 Schottky -> VSYS (Pico). USB 5V can not back-feed the batteries. Never use 4 cells (Pico VSYS max 5.5V).

1. POWER



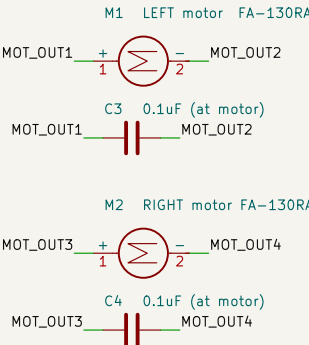
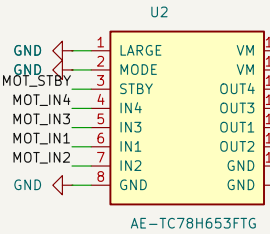
Insert the plug first and solder the header so that the RED wire is on '+'

MECHANICAL: H1-H4 = gearbox frame M3 (x30/x70, y62/y96), H5 = front skid M3 (x50, y11: behind the centre sensor)



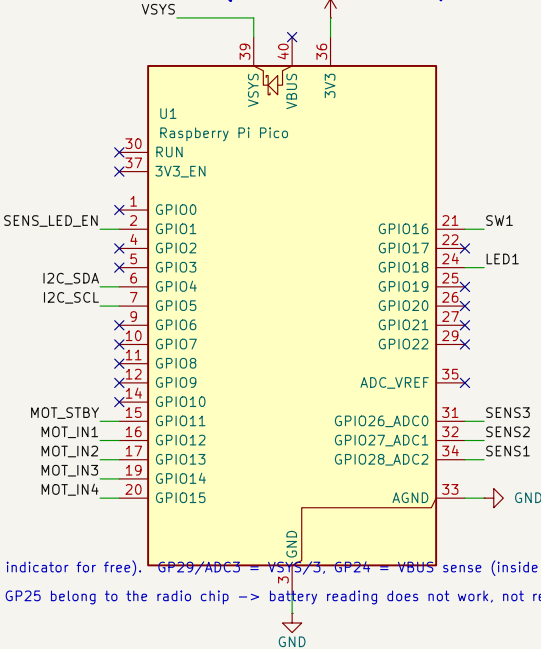
Battery voltage: no divider on this board. The Pico measures VSYS/3 on GP29 (ADC3) internally:
VBAT = VSYS + about 0.3 V (D1). GP24 = 1 while USB is plugged in.

3. MOTOR DRIVER (Akizuki AE-TC78H653FTG module, DIP16)



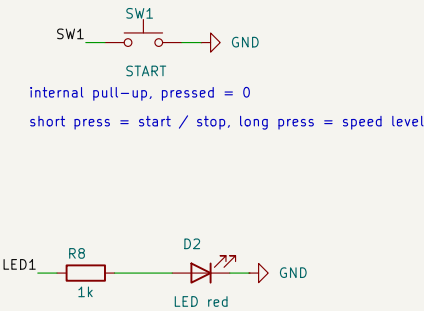
LARGE=L : small mode (2 motors, 2A/ch) MODE=L : IN/IN control
IN1/IN2 -> OUT1/OUT2 = LEFT motor, IN3/IN4 -> OUT3/OUT4 = RIGHT motor
H/L fwd, L/H rev, H/H brake, L/L coast. Over-current (ISD) latches OFF:
toggle STBY L->H to recover (done by the library).

2. MCU Raspberry Pi Pico / Pico 2 (soldered flat, no pin headers)



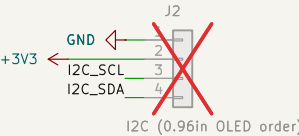
GP25 = on-board LED (a 2nd indicator for free). GP29/ADC3 = VSYS/3, GP24 = VBUS sense (inside Pico).
Pico W / Pico 2 W: GP29 and GP25 belong to the radio chip -> battery reading does not work, not recommended.

4. BUTTON / LED



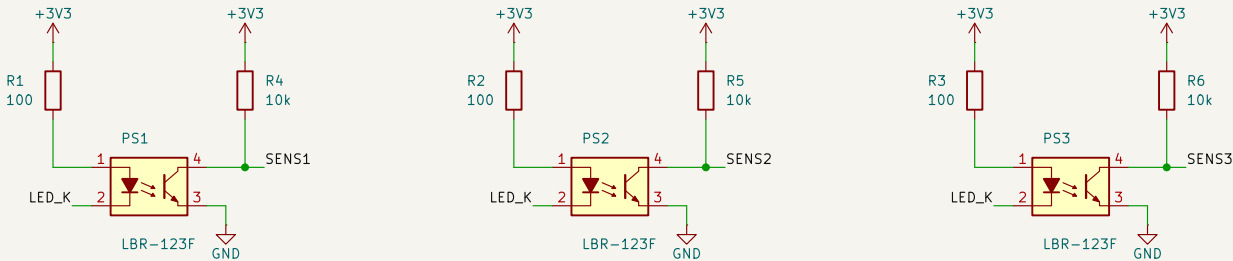
no buzzer, no power LED on the Lite board

5. EXPANSION (holes only)

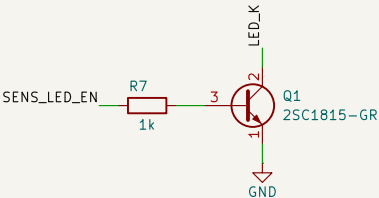


6. LINE SENSORS 3 x LBR-123F (mount on the BOTTOM side, lens to the floor, 2-3 mm gap), 12 mm apart

IR LED: (3.3V - 1.25V) / 100 ohm = about 19 mA each. Q1 switches all LEDs (ambient light cancel). Photo transistor: 10k pull-up -> white = low voltage, black = high voltage



PS1 = LEFT -> GP28/ADC2, PS2 = CENTRE -> GP27/ADC1, PS3 = RIGHT -> GP26/ADC0 (seen from above, robot moving up the page)



Q1: all IR LEDs on when SENS_LED_EN = 1 (about 57 mA)

Design notes: lite/README.md
Pico + Akizuki AE-TC78H653FTG module + 3 x LBR-123F
All parts through-hole. Pico soldered flat (castellated edge)

EDTC
Sheet: /
File: Linetracer2-Lite.kicad_sch

Title: Linetracer2 Lite - low-cost kids line tracer

Size: A3 Date: 2026-10-05
KiCad E.D.A. 10.0.6

Rev: L1
Id: 1/1